



Module Presenter's Manual

for

Large Data Management (MongoDB)

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Ver. 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Master the skills of Replication and Sharding of data in MongoDB to optimize read/write performance
- Perform installation, configuration, and maintenance of MongoDB environment
- Get hands-on experience in creating and managing different types of indexes in MongoDB for query execution
- Develop skill sets in processing huge amounts of data using MongoDB tools
- Proficiently store unstructured data in MongoDB
- Gain proficiency in MongoDB configuration, backup methods as well as monitoring and operational strategies
- Acquire in-depth understanding of managing DB Notes, replica set & Master-Slave concepts

2. Information on Session Allocation

Module	Online Hours	Self-study (No. of Hours)
Large Data Management (MongoDB)	32	10

Throughout this Presenter's Manual, the module, **Large Data Management (MongoDB)**, will be referred to as **MongoDB**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables

Resources available on Onlinevarsity:

Feature - Description/Functionality
Ebook -Student reading material in the form of PDF
Download eBook - Student has the option to download the subject related e-book and read offline
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3		Day 4	
1	Session 1 MONGODB – TL1	Session 2 MONGODB – TL2	Session 3 MONGODB –TL3		Session 4 MONGODB – TL4	
2	Session 5 MONGODB –TL5	Session 6 MONGODB – TL6	Session 7 MONGODB – TL7	S1-S3	Session 8 MONGODB –TL8	
3	Session 9 MONGODB – TL9	Session 10 MONGODB – TL10	Session 11 MONGODB –TL11		Session 12 MONGODB – TL12	
4	Session 13 MONGODB – TL13	Session 14 MONGODB –TL14	Session 15 MONGODB – TL15		Session 16 MONGODB – TL16	S4-S5

MONGODB-Large Data Management (MongoDB)

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	MONGODB – TL1	All the topics as listed below from Session 1 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 1 - Introduction to MongoDB</u> - Databases - An Overview of Types of DBMS - RDBMS VS NoSQL - Overview of MongoDB - What is MongoDB and how it works - Database, Collection, and Document - Getting started with MongoDB - MongoDB Server - MongoDB Cloud - MongoDB tools - Install MongoDB Shell - Install MongoDB Compass - Connecting the server with MongoDBShell	Managing Large Databases Using MongoDB SG-Session 1 XP-Session 1 TG-Session 1
2	MONGODB – TL2	All the topics as listed below from Session 2 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 2 - Working with MongoDB Databases</u> - Data Types in MongoDB - Databases and Collections - Create Database - Create Collection - Insert Document - Query Document - Update Document - Delete Document - Drop Collection - Drop Database	Managing Large Databases Using MongoDB SG-Session 2 XP-Session 2 TG-Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
3	MONGODB – TL3	All the topics as listed below from Session 3 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 3 - MongoDB Operators</u> • Introduction to MongoDB operators • Query and Projection Operator • Comparison • Logical • Element • Array • Projection • Miscellaneous • Update Operator • Field • Array	Managing Large Databases Using MongoDB SG-Session 3 XP-Session 3 TG-Session 3
4	MONGODB – TL4	All the topics as listed below from Session 4 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 4 – Aggregation Pipeline</u> • Introduction to Aggregation Pipeline • Aggregation Pipeline Stages • Aggregation Expressions	Managing Large Databases Using MongoDB SG-Session 4 XP-Session 4 TG-Session 4
5	MONGODB – TL5	All the topics as listed below from Session 5 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 5 – Database Commands</u> • Introduction to Database Commands • Authentication commands • Write operation commands • Geospatial commands • Aggregation commands	Managing Large Databases Using MongoDB SG-Session 5 XP-Session 5 TG-Session 5
6	MONGODB – TL6	All the topics as listed below from Session 6 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 6 – MongoDB Shell Methods</u>	Managing Large Databases Using MongoDB

Session No.	Session Title	Session Details	Deliverables' Mapping
		• Introduction to MongoDB Shell methods • Collection methods • Database methods • User management methods • Role management methods	SG-Session 6 XP-Session 6 TG-Session 6
7	MONGODB – TL7	The Try It Yourself questions of Session 1 to Session 6 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session.	Managing Large Databases Using MongoDB Session 1-6
	S1-S3	All the Assignment topics from Session 1 to Session 6 of <i>Managing Large Databases Using MongoDB</i> course should be covered in this session.	<u>Managing Large Databases Using MongoDB</u> (OnlineVarsity) Session 1-6
8	MONGODB – TL8	All the topics as listed below from Session 7 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 7 – MongoDB Indexing</u> • Introduction to Indexes • Multi-key Index • Hashed Index • Create Index • Drop Index	Managing Large Databases Using MongoDB SG-Session 7 XP-Session 7 TG-Session 7
9	MONGODB – TL9	All the topics as listed below from Session 8 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 8 – Replication and Sharding</u> • Replication • What is replication? • Deployment Architecture • Replication Commands and Methods • Implement Replication Using an Example	Managing Large Databases Using MongoDB SG-Session 8 XP-Session 8 TG-Session 8
10	MONGODB – TL10	All the topics as listed below from Session 8 of <i>Managing Large Databases Using</i>	Managing Large Databases Using MongoDB

Session No.	Session Title	Session Details	Deliverables' Mapping
		MongoDB book should be covered in this session. <u>Session 8 – Replication and Sharding</u> - Sharding - What is Sharding? - Cluster Components - Shard Keys - Sharding Commands and Methods Implement Sharding Using an Example	SG-Session 8 XP-Session 8 TG-Session 8
11	MONGODB – TL11	All the topics as listed below from Session 9 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 9 – Transaction Management</u> - Introduction to MongoDB Transactions - Transaction Properties - Transactions and Sessions - Working With Transactions	Managing Large Databases Using MongoDB SG-Session 9 XP-Session 9 TG-Session 9
12	MONGODB – TL12	All the topics as listed below from Session 10 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 10 – MongoDB Tools</u> - Connect MongoDB Compass with MongoDB deployment - Manage Database Using Compass - Install MongoDB BI Connector	Managing Large Databases Using MongoDB SG-Session 10 XP-Session 10 TG-Session 10
13	MONGODB – TL13	All the topics as listed below from Session 10 of <i>Managing Large Databases Using MongoDB</i> book should be covered in this session. <u>Session 10 – MongoDB Tools</u> - Connect MongoDB Compass with MongoDB deployment - Manage Database Using Compass - Install MongoDB BI Connector	Managing Large Databases Using MongoDB SG-Session 10 XP-Session 10 TG-Session 10

Session No.	Session Title	Session Details	Deliverables' Mapping
14	MONGODB – TL14	All the topics as listed below from Session 11 and Session 12 of Managing Large Databases Using MongoDB book should be covered in this session. <u>Session 11 – MongoDB Cloud</u> • Get Started with Atlas • Access MongoDB Atlas Cluster • Import Data into Atlas Cluster From MongoDB Instance • Export Data from Atlas Cluster Into MongoDB Instance • Administer MongoDB Cluster	Managing Large Databases Using MongoDB SG-Session 11 XP-Session 11 TG-Session 11
15	MONGODB – TL15	All the topics as listed below from Session 11 and Session 12 of Managing Large Databases Using MongoDB book should be covered in this session. <u>Session 12 – MongoDB Database Connectivity with Python</u> • Overview of connecting Python with MongoDB • Install PyMongo driver • Create a Database and Collection Using Python • Insert Documents Using Python • Query Documents Using Python	Managing Large Databases Using MongoDB SG-Session 12 XP-Session 12 TG-Session 12
16	MONGODB – TL16	The Try It Yourself questions of Session 10 to Session 12 of Managing Large Databases Using MongoDB book should be covered in this session.	Managing Large Databases Using MongoDB Session 10-12
	S4-S5	All the Assignment topics from Session 10 to Session 12 of Managing Large Databases Using MongoDB course should be covered in this session.	<u>Managing Large Databases Using MongoDB</u> (OnlineVarsity) Session 10-12

6. Library References

➤	MongoDB: The Definitive Guide by Kristina Chodorow and Michael Dirolf
➤	Mastering MongoDB 4.x by Wouter Thielen
➤	Data Modeling for MongoDB: Building Well-Designed and Supportable MongoDB Databases by Steve Hoberman, William Steve Hoberman

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